

Varadhan-type asymptotics for boundary data with localized support

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Let $A = A(x) \in \mathbb{R}^{N \times N}$, $N \geq 2$, be a symmetric matrix and consider the linear operator

$$\mathcal{L}_A u(x) = \operatorname{tr} \left(A(x) \nabla^2 u(x) \right),$$

where $A \in C^1(\mathbb{R}^N)$ is such that $\mathcal{L}_A$ is **uniformly elliptic**.

- $A(x) \equiv I_d$ then $\mathcal{L}_A = \Delta$;
- $\langle \cdot, \cdot \rangle_A := \langle A^{-1} \cdot, \cdot \rangle$ induces a distance $d = d_A$ for every $x, y \in \mathbb{R}^N$:

$$d(x, y) = \inf \left\{ \int_0^T \|\dot{\gamma}(t)\|_A dt : \gamma \in AC(0, T), \gamma(0) = x, \gamma(T) = y \right\};$$

- $\Omega \subset \mathbb{R}^N$ shall be a bounded domain (sufficiently smooth) with boundary $\partial\Omega$;
- we denote by

$$d_{\partial\Omega}(x) = \inf_{y \in \partial\Omega} d(x, y).$$

- Varadhan (1967) studied, in a domain $\Omega \subset \mathbb{R}^N$, the problems

$$U_\varepsilon = U_\varepsilon(x) :$$

$$\begin{cases} U_\varepsilon - \varepsilon^2 \mathcal{L}_A U_\varepsilon = 0 & \text{in } \Omega, \\ U_\varepsilon = 1 & \text{on } \partial\Omega, \end{cases}$$

$$\varepsilon > 0;$$

$$u = u(x, t) :$$

$$\begin{cases} u_t - \mathcal{L}_A u = 0 & \text{in } \Omega \times (0, \infty), \\ u = 0 & \text{on } \Omega \times \{0\}, \\ u = 1 & \text{on } \partial\Omega \times (0, \infty). \end{cases}$$

Note, by the *Laplace transform*

$$U_\varepsilon(x) = \int_0^\infty e^{-\tau/\varepsilon^2} du(x, \tau) \text{ for every } x \in \Omega, \varepsilon > 0.$$

Varadhan formulas

For every $x \in \overline{\Omega}$,

$$\lim_{\varepsilon \rightarrow 0^+} [-\varepsilon \log U_\varepsilon(x)] = d_{\partial\Omega}(x),$$

and

$$\lim_{t \rightarrow 0^+} [-4t \log u(x, t)] = d_{\partial\Omega}(x)^2.$$

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(Brief) recap of related results

- [Evans–Ishii (1985)] proves Varadhan formulas in the more general context of **vanishing viscosity method** for viscosity solutions (and in relation with optimal control theory; see also [Fleming]).
- **Extensions to various nonlinear settings:** the porous medium equation, the evolutionary p -Laplace equation, fast diffusion equations [Magnanini-Sakaguchi (2000–)]; the game-theoretic p -Laplacian, and the Pucci operators [B.-Magnanini (2019-20-22)].
- In particular, [B.-Magnanini] also establish the convergence rates $O(\varepsilon \log \varepsilon)$ as $\varepsilon \rightarrow 0^+$ and $O(t \log t)$ as $t \rightarrow 0^+$.

Non-constant boundary values

- It is natural to consider the more general Dirichlet or Cauchy–Dirichlet problems with **non-constant data** $f : \partial\Omega \rightarrow \mathbb{R}$:

$$U_\varepsilon = U_\varepsilon(x):$$

$$\begin{cases} U_\varepsilon - \varepsilon^2 \mathcal{L}_A U_\varepsilon = 0 & \text{in } \Omega, \\ U_\varepsilon = f & \text{on } \partial\Omega, \end{cases}$$

$$\text{for } \varepsilon > 0,$$

and

$$u = u(x, t):$$

$$\begin{cases} u_t - \mathcal{L}_A u = 0 & \text{in } \Omega \times (0, \infty), \\ u = 0 & \text{on } \Omega \times \{0\}, \\ u = f & \text{on } \partial\Omega \times (0, \infty). \end{cases}$$

Our contribution: in B.-Magnanini-Marini, *Short-time behaviour and invariant surfaces for caloric functions with non-constant boundary values*, in preparation (2026), in the case of variable f ,

for both the elliptic and parabolic problems, we obtain:

- Extensions of Varadhan formulas;
- Extensions of some formulas for means on spheres;
- Rigidity results under short-time and slow-diffusion invariances.

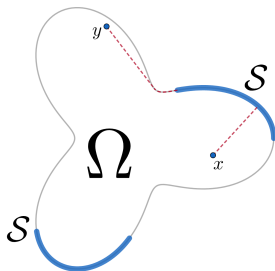
(Other) notations

Case $f \geq 0$. Let $f \in C(\partial\Omega)$ be non-negative (not identically zero), and set

$$\mathcal{S} = \{x \in \partial\Omega : f(x) > 0\}.$$

We define the **intrinsic distance (inside Ω) from a point x to $\mathcal{S}$** :

$$d_{\mathcal{S}}(x) := \inf \left\{ \int_0^T \|\dot{\gamma}(t)\|_A : \gamma \in AC(0, T; \overline{\Omega}), \gamma(0) = x, \gamma(T) \in \mathcal{S} \right\}.$$



In **dashed red**, the shortest paths defining $d_{\mathcal{S}}$ from x and y .

Theorem 1 (non-negative f)

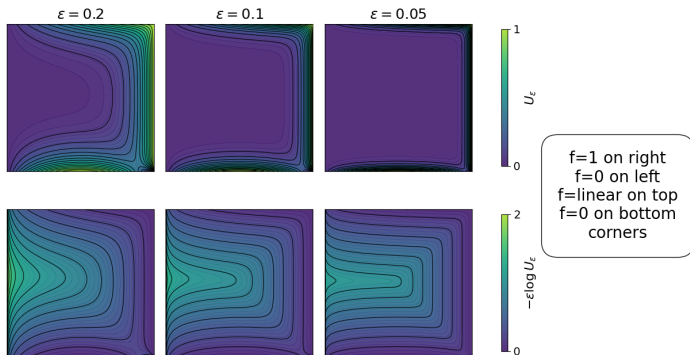
$\Omega \subset \mathbb{R}^N$ bounded with $\partial\Omega \in C^{1,1}$. Then, for every $x \in \Omega \cup \mathcal{S}$:

$$\lim_{\varepsilon \rightarrow 0^+} [-\varepsilon \log U_\varepsilon(x)] = d_{\mathcal{S}}(x),$$

and

$$\lim_{t \rightarrow 0^+} [-4t \log u(x, t)] = d_{\mathcal{S}}(x)^2,$$

uniformly on compact subsets of $\Omega \cup \mathcal{S}$.



Main tools

- Reduction to the case $f = \mathcal{X}_S$.
- Mixed **Varadhan** and **Evans-Ishii** approach: first we study the elliptic formula, in the context of vanishing viscosity, and then we deduce the parabolic one.

Main difficulties

- No fine global barriers but *only* local barriers **near $\partial\Omega \setminus \partial S$ for the case $f = \mathcal{X}_S$.**
- By Hopf-Cole transform, " $V_\varepsilon := -\varepsilon \log U_\varepsilon \rightarrow V_0$ "

$$\begin{cases} -\varepsilon \mathcal{L}_A V_\varepsilon + \langle A \nabla V_\varepsilon, \nabla V_\varepsilon \rangle = 1 & \text{in } \Omega \\ V_\varepsilon = -\varepsilon \log f & \text{on } S \\ \limsup_{x \rightarrow y_0} V_\varepsilon(x) \rightarrow +\infty & \text{for } y_0 \in \partial\Omega \setminus S. \end{cases} \Rightarrow \begin{cases} \langle A \nabla V_0, \nabla V_0 \rangle = 1 & \text{in } \Omega, \\ V_0 = 0 & \text{on } S, \\ V_0 = ? & \text{on } \partial\Omega \setminus S; \end{cases}$$

Goal: to prove that **vanishing viscosity** selects d_S (for instance, d_S is the maximal viscosity solution of $\langle A \nabla V_0, \nabla V_0 \rangle = 1$ in Ω and $V_0 = 0$ on S).

The case of sign-changing f

Let $f \in C(\partial\Omega)$ **change sign**; define

$$\mathcal{S}^+ = \{x \in \partial\Omega : f(x) > 0\}, \quad \mathcal{S}^- = \{x \in \partial\Omega : f(x) < 0\},$$

set $\mathcal{S} = \mathcal{S}^+ \cup \mathcal{S}^-$, and the **set of points equidistant from $\mathcal{S}^+$ and $\mathcal{S}^-$**

$$\mathcal{G} = \{x \in \Omega \cup \mathcal{S} : d_{\mathcal{S}^+}(x) = d_{\mathcal{S}^-}(x)\}.$$

Corollary 2 (sign-changing f)

For every $x \in \Omega \cup \mathcal{S} \setminus \mathcal{G}$, we get

$$\lim_{\varepsilon \rightarrow 0^+} [-\varepsilon \log |U_\varepsilon(x)|] = d_{\mathcal{S}}(x),$$

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What about the formulas in $\mathcal{G}$?

If $x \in \mathcal{G}$:

$U_\varepsilon(x) = 0$ for every $\varepsilon > 0$ or $U_\varepsilon(x)$ oscillates around 0, for $\varepsilon \rightarrow 0^+ \Rightarrow$
 $\limsup_{\varepsilon \rightarrow 0^+} [-\varepsilon \log U_\varepsilon(x)] = \infty.$

👁 **Symmetric example:** $A(x) \equiv I_d$, $\Omega = B(0)$, $f(x', x_N) = -f(x', x_N) \Rightarrow$
 $\{U_\varepsilon = 0\} = \{x \in \Omega : x_N = 0\} \subset \mathcal{G}.$

$|U_\varepsilon(x)| > 0$, for $0 < \varepsilon \leq \varepsilon^* \Rightarrow$ open question to find sufficient condition for Corollary 2 to hold.

👁 **Positive example:** $A \equiv I_d$, $\Omega = B_1(0)$ and $f(x) = \sum_{j=1}^m c_j \mathcal{Y}_{k_j}(x/|x|)$, for $\mathcal{Y}_k$ spherical harmonics: asymptotic expansion in terms of Bessel functions yields the Varadhan formulas throughout $\Omega \cup S$, **including every point of $\mathcal{G}$.**

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What about the size of $\mathcal{G}$?

- “The best case” is that $\text{int}(\mathcal{G}) = \emptyset$;

Sufficient conditions for $\text{int}(\mathcal{G}) = \emptyset$

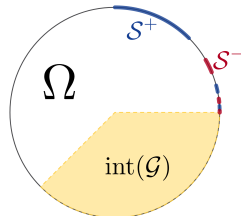
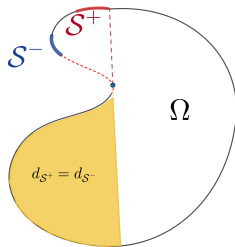
- $\overline{\mathcal{S}} = \partial\Omega$;
 - Ω convex and $\overline{\mathcal{S}^+} \cap \overline{\mathcal{S}^-} \subset \text{int}(\mathcal{S})$.
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- It is easy to build examples in which $\text{int}(\mathcal{G}) \neq \emptyset$:

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- $\overline{\mathcal{S}} = \partial\Omega$;
 - Ω convex and $\overline{\mathcal{S}^+} \cap \overline{\mathcal{S}^-} \subset \text{int}(\mathcal{S})$.
- It is easy to build examples in which $\text{int}(\mathcal{G}) \neq \emptyset$:



- **Varadhan formulas** have applications in the study of **rigidity results** for proper **overdetermined** problems for both U_ε or $u(x, t)$.

- A $(N - 1)$ -surface Γ is called
 - a **diffusion-invariant surface** if

$$U_\varepsilon(x) = c_\varepsilon \quad \text{for every } x \in \Gamma,$$

where $c = c_\varepsilon$ is a function defined in a subset of values $\varepsilon \in I \subset (0, \infty)$.

- a **time-invariant surface** if

$$u(x, t) = c(t) \quad \text{for every } x \in \Gamma,$$

where $c = c(t)$ is a function defined in a set of times $t \in I \subset (0, \infty)$.

A classical rigidity result

- From now on, $A(x) \equiv I_d$.

A main result when $f \equiv 1$, from Magnanini-Sakaguchi (2002):

Magnanini-Sakaguchi rigidity result

$\Omega \subset \mathbb{R}^N$, $N \geq 2$, bounded; if Ω has the exterior sphere condition and there exists a time-i.s.

$$\Gamma = \partial D \quad \text{with} \quad \overline{D} \subset \Omega,$$

where D satisfies the interior cone condition.

Then, Ω **must be a ball (and the solution radial)**.

The key point:

- Varadhan formulas $\Rightarrow \Gamma = \{d_{\partial\Omega} = R\}$.

A strategy (from Magnanini-Sakaguchi (2002)):

- formulas for the mean of spheres + balance laws $\Rightarrow$ constraint on the principal curvatures of Γ and $\partial\Omega$.

Asymptotics for the mean value on tangent spheres

Theorem 3

Suppose that $f \in C(\partial\Omega)$, $f \geq 0$. We have:

$$\lim_{\varepsilon \rightarrow 0^+} \varepsilon^{-\frac{N-1}{2}} \int_{\partial B_R(p)} U_\varepsilon(x) dS_x = f(q) C_N \left\{ \prod_{j=1}^{N-1} [1/R - \kappa_j(q)] \right\}^{-1/2},$$

and

$$\lim_{t \rightarrow 0^+} t^{-\frac{N-1}{4}} \int_{\partial B_R(p)} u(x, t) dS_x = f(q) C'_N \left\{ \prod_{j=1}^{N-1} [1/R - \kappa_j(q)] \right\}^{-1/2}.$$

Here:

$$B_R(p) \subset \Omega, \quad \partial B_R(p) \cap \partial\Omega = \{q\},$$

and

$$\kappa_j(q) < 1/R \quad \text{for} \quad j = 1, \dots, N-1,$$

$\kappa_1(q), \dots, \kappa_{N-1}(q)$ = the principal curvatures of $\partial\Omega$ at q , with respect to the interior normal.

Rigidity results: $f \geq 0$

Let $\Omega \subset \mathbb{R}^N$, $N \geq 2$, bounded.

Assume either

- (a) Ω satisfies the exterior sphere condition and $\mathcal{S} = \partial\Omega$; or
- (b) $\Omega \in C^{1,1}$ and $\bar{\mathcal{S}} = \partial\Omega$.

Theorem 4 [Rigidity under **two** invariant surfaces]

Suppose that, for some domains satisfying the interior cone condition D_1, D_2 , with $\overline{D_1}, \overline{D_2} \subset \Omega$, we have

$$U_\varepsilon(x) = c_\varepsilon^i \quad \text{for } x \in \partial D_i \quad \text{and} \quad 0 < \varepsilon \leq \varepsilon^*, \quad \text{for } i = 1, 2,$$

or

$$u(x, t) = c_i(t) \quad \text{for } x \in \partial D_i, \quad 0 < t \leq t^*, \quad \text{for } i = 1, 2,$$

provided that $\partial D_1 \cap \partial D_2 = \emptyset$.

Then, **f is constant**

(and hence Ω is a ball and the solutions are radial).

- Investigate the rigidity results **without the assumption** $f \geq 0$.

Expected result

Let Ω be sufficiently smooth and either

- (a) $d_S = d_{\partial\Omega}$;
- (b) Ω convex and $\text{int}(\mathcal{G}) = \emptyset$;

Let $\overline{D} \subset \Omega$ be such that either

$$U_\varepsilon(x) = c_\varepsilon \quad \text{for } x \in \partial D \quad \text{and } 0 < \varepsilon \leq \varepsilon^*,$$

or

$$u(x, t) = c(t) \quad \text{for } x \in \partial D, \quad 0 < t \leq t^*.$$

Then, f **must have constant sign**.

Thank you for your attention

- If $f \leq 0$, the same formulas hold for $-U_\varepsilon$ and $-u(x, t)$.
- When $f > 0$, the proof is a direct consequence of Varadhan original results:

$$\left(\min_{\partial\Omega} f\right) w(x, t) \leq u(x, t) \leq \left(\max_{\partial\Omega} f\right) w(x, t) \quad \text{where} \quad w \equiv 1 \quad \text{on} \quad \partial\Omega,$$

by comparison principle.

- When $\overline{\mathcal{S}} = \partial\Omega$, we recover the same Varadhan formulas, apart from the points belonging to $\partial\Omega \setminus \mathcal{S}$, where “ $-\varepsilon \log U_\varepsilon = \infty$ ” and “ $-4t \log u = \infty$ ”.

Non-constant boundary values

- It is natural to consider the more general Dirichlet or Cauchy–Dirichlet problems with **non-constant data**:

$$U_\varepsilon = U_\varepsilon(x):$$

$$\begin{cases} U_\varepsilon - \varepsilon^2 \mathcal{L}_A U_\varepsilon = u_0 & \text{in } \Omega, \\ U_\varepsilon = f & \text{on } \partial\Omega, \end{cases} \quad (E_{f,u_0})$$

$$\text{for } \varepsilon > 0,$$

and

$$u = u(x, t):$$

$$\begin{cases} u_t - \mathcal{L}_A u = 0 & \text{in } \Omega \times (0, \infty), \\ u = u_0 & \text{on } \Omega \times \{0\}, \\ u = f & \text{on } \partial\Omega \times (0, \infty), \end{cases} \quad (P_{f,u_0})$$

where $u_0 \in L^2(\Omega)$, and $f : \partial\Omega \rightarrow \mathbb{R}$.

Case $u_0 \not\equiv 0$

. When $A(x) \equiv I_d$, in [Hasebe, Masamune, Oka, Sakai, Yamada (preprint 2024)], it is assumed

$$u_0 \geq 0, \quad \overline{E} \subset \Omega, \quad E := \{u_0 = 0\}, \quad \text{and} \quad f \geq 0$$

and it is proved:

$$\lim_{\varepsilon \rightarrow 0^+} [-\varepsilon \log U_\varepsilon(x)] = d_{\partial E}(x) \quad \text{for } x \in \overline{E}.$$

👁 the formula **does not** depend on the boundary value f .

Expected result 2

Suppose that either $\text{int}(\mathcal{G}) = \emptyset$ or $\overline{\mathcal{S}} = \partial\Omega$.

Suppose Γ is a small-diffusion (or short-time) invariant surface (sufficiently smooth) such that $\Gamma \cap \partial\Omega \neq \emptyset$.

Then, Γ is a ruled surface.